# Evaluating Agentic AI for Supply-Chain Disruption Mitigation

A Two-Phase Paired-Simulation Study of a Bounded LLM Agent Against a Classical Heuristic

*Research Study – GitHub Repository, Live Demo*

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## Abstract

Can a bounded, tool-using large language model (LLM) agent mitigate supply-chain disruption more effectively than a transparent, cost-minimizing classical heuristic – and if so, under what conditions? We answer this in two phases, sharing one paired-simulation infrastructure throughout. **Phase 1** tested one network topology against three disruption severities (300 real, live-API replications) and found the LLM agent losing under Light disruption but winning decisively under Medium/Heavy – with every one of 300 replications agreeing unanimously (100%/0% per profile), a result too clean to trust at face value. Auditing why traced the unanimity to insufficient environmental randomness and a single, low-redundancy topology, not to a genuine effect ceiling. **Phase 2** redesigned the experiment along exactly the axes the audit implicated: three network topologies crossed with the same three severities (nine cells, 498 replications), plus per-replication randomness in disruption timing, duration, disclosure, and shipment volume. The redesign both validated Phase 1 (the original topology reproduces its original pattern almost exactly) and overturned its implicit generality: **network topology does not merely scale the LLM's advantage – it can reverse its sign**, with the highest-redundancy topology favoring the heuristic ever more strongly as severity rises, the opposite of the other two topologies. Across the whole two-phase program, agentic AI shows a real, statistically confirmed advantage for high-stakes disruption response on constrained networks, and a real, equally confirmed disadvantage under low-stakes conditions and high-redundancy networks – a nuanced, conditional answer, not a universal one. This edition adds the full-grid heatmap and the signal-to-noise validation figure, plus a follow-up section covering both nearer-term scope extensions and a reinforcement-learning formulation of the same decision problem.


## 1 Research Problem

Agentic AI – LLM-based systems that reason, call tools, and act – is increasingly proposed for operational supply-chain decisions: rerouting, expediting, or cancelling shipments in response to disruption, where a wrong call has directly measurable cost. The claim that "AI beats the baseline" rarely survives scrutiny in this domain: mismatched scenarios, cherry-picked examples, or a classical baseline never given a fair, symmetric shot. This study's overarching goal, spanning both phases reported here, is to test that claim rigorously for one narrow, well-defined decision – *what should happen to a shipment sitting at a node when the surrounding network is disrupted* – and to answer not just *whether* an agentic policy helps, but *when*: **under identical disrupted conditions and identical operational constraints, can a bounded LLM policy reduce the incremental cost of disruption more effectively and more robustly than a transparent classical heuristic, and how does that advantage depend on network structure and disruption severity?**

Fairness is operationalized structurally, not asserted: both policies observe identical read-only state, are validated and fall back through the same machinery, and face the same stochastic demand, delay, and disruption realization within a replication. Each policy is scored against its own undisrupted counterfactual, isolating the *incremental* cost of disruption rather than baseline operating cost.

## 2 Related Work

**LLM agents with tools.** ReAct [1] established the now-standard interleaved reasoning–action–observation loop for LLM agents, later made reliable via structured function calling; Toolformer [2] showed LLMs can learn *when* to invoke tools rather than merely being prompted to. Our agent follows this lineage: five deterministic, read-only tools, an 8-call budget, temperature 0, and a hard fallback to the classical heuristic on abstention or invalid output.

**LLM agents in supply chain management.** This is an active, recent direction: InvAgent [3] applies multi-agent LLMs to inventory management under demand uncertainty; broader surveys [4] catalog knowledge-integration and negotiation as dominant use cases; [5] studies autonomous multi-agent consensus across supply-chain roles. This literature consistently flags a verification gap – agentic LLMs can inherit biases and produce confidently wrong decisions absent explicit checking – which is precisely why this study reports null and unfavorable results for the agent alongside favorable ones, rather than only the latter.

**Classical disruption management.** The heuristic baseline sits in a long tradition of cost-minimizing routing and recovery rules under disruption [6, 7]: transparent, auditable, structurally incapable of considering information outside its candidate-route cost estimate. It is not a strawman – it wins cleanly in both phases under Light disruption, and dominates one topology entirely in Phase 2.

| Severity | Mean Δ | 95% CI | LLM win |
|---|---|---|---|
| Light | +992 | [967, 1018] | 0% |
| Medium | −3,911 | [−4011, −3811] | 100% |
| Heavy | −74,490 | [−75919, −73061] | 100% |

**Table 1:** Phase 1: mean Δ and 95% CI, $n = 100$/severity. Every one of the 300 replications agreed unanimously with every other in its profile.

# 3 Shared Methodology

Both phases run on one unmodified simulator: a discrete-time, single-product network (suppliers → ports → hubs → plant) in which a policy intervenes on at most one shipment per decision point with exactly one action – `WAIT`, `REROUTE`, `EXPEDITE`, or `ABSTAIN` – validated by shared rules with a heuristic-then-`WAIT` fallback on failure.

One random seed drives one *replication*: demand, ordinary transport delay, and (Phase 2 only) disruption realization and shipment volume are drawn entirely before either policy acts. Four cloned branches then run per replication – {heuristic, LLM} × {undisrupted, disrupted} – sharing an identical 20-day warm-up snapshot. Each day's cost sums seven components:

$$J = C_{\text{transport}} + C_{\text{reroute}} + C_{\text{expedite}} + C_{\text{holding}} + C_{\text{backlog}} + C_{\text{late}} + C_{\text{terminal}} \tag{1}$$

Per policy $p$, the disruption's incremental cost is:

$$\text{TCD}_p = J_p^{\text{disrupted}} - J_p^{\text{undisrupted}} \tag{2}$$

and the reported effect compares the two policies directly:

$$\Delta = \text{TCD}_{\text{LLM}} - \text{TCD}_{\text{heuristic}} \tag{3}$$

with $\Delta < 0$ favoring the LLM. Pairing removes common-mode noise, since the same random world hits both policies, giving far more power per replication than an unpaired design.

# 4 Phase 1: Initial Evaluation

Phase 1 tested one network (5 nodes, 6 edges, ∼3 end-to-end routes) against three fixed disruption severities – a partial-capacity cut (Light), a full 7-day port closure (Medium), and a 3-week closure plus alternate-route congestion (Heavy) – 100 live-API replications each, 300 total.

**Headline result.** The LLM agent loses under Light disruption but wins decisively under Medium and Heavy (Table 1) – a clean, seemingly interpretable severity gradient.

**The problem:** within every profile, all 100 replications agreed with each other exactly (100%/0%), never once split. Auditing why traced this to three compounding, fixable design properties, not a genuine effect ceiling: (i) the shipment release schedule was fully deterministic, exactly 40 units every day; (ii) the disruption's timing, duration, and disclosure were fully known in advance; (iii) the network offered only ∼3 routes – not enough structure for a close call to ever arise. Quantitatively, the mean $|\Delta|$ was 7.6×–10.2× larger than the between-replication standard deviation of Δ in every profile: the experiment could not generate a world where the effect and the noise were remotely comparable in scale, so unanimity was close to guaranteed by construction, not by an overwhelming true effect. This motivated Phase 2.

| Topology | Severity | Mean Δ | 95% CI | LLM win |
|---|---|---|---|---|
| Compact | Light | +935 | [814, 1055] | 0% |
| Compact | Medium | −1,436 | [−2210, −663] | **75.8%** |
| Compact | Heavy | −56,915 | [−68896, −44934] | 100% |
| Standard | Light | +921 | [857, 985] | 0% |
| Standard | Medium | −4,192 | [−4682, −3702] | 100% |
| Standard | Heavy | −111,850 | [−122960, −100741] | 100% |
| Extended | Light | +1,636 | [1451, 1820] | 0% |
| Extended | Medium | +5,176 | [4536, 5817] | 0% |
| Extended | Heavy | +22,114 | [17950, 26279] | 0% |

**Table 2:** Phase 2: mean Δ and 95% CI on all nine cells. $n = 33$ (Compact/Extended), $n = 100$ (Standard). Every CI excludes zero.

# 5 Phase 2: Topology-Generalized Evaluation

Phase 2 targets the audit's three causes directly, without altering the shared simulator, action space, validator, cost model, or pairing logic (Section 3).

**Topology:** three tiers – Compact (4 nodes/4 edges/0 non-emergency reroutes), Standard (Phase 1's original network, byte-identical), and Extended (10 nodes/16 edges/7 structurally distinct non-emergency routes, confirmed by betweenness centrality) – crossed independently with the same three severities.

**Randomness:** disruption start day, duration, and disclosure delay are now sampled per replication rather than fixed (a zero-uncertainty template provably reproduces Phase 1 bit-for-bit); shipment quantity is sampled per release on Compact/Extended (Standard keeps Phase 1's fixed 40 units by design, as an unmodified behavioral anchor). Extended's severity scenarios target `hub_1` rather than `port_primary`, since centrality analysis showed the latter is no longer that tier's structural chokepoint once seven alternate routes exist.

Nine cells result, each an independently-run configuration. Compact and Extended run 33 replications/cell; Standard runs Phase 1's original 100/cell as the reference column. All nine were calibrated (3 replications each) before the full run, which then executed live against the OpenAI API: 498 replications, ∼14,700 LLM calls, ∼4.5 h wall-clock across three concurrent per-topology processes, all nine completing without error.

**Validation.** Standard × Medium is Phase 1's original configuration, re-run unmodified at Phase 2 scale: LLM wins 100% of 100 replications, matching Phase 1's original pattern almost exactly (compare Table 1's Medium row) – the shared infrastructure did not silently break.

**Headline result:** Compact and Standard reproduce Phase 1's shape, the LLM's advantage growing with severity.

**Extended inverts it:** the heuristic's advantage grows with severity instead, winning every replication at every severity (Figure 1). Topology is not a magnitude dial on a fixed direction of effect; it can flip which policy is favored as conditions worsen. This is visible at a glance across the whole grid in Fig-

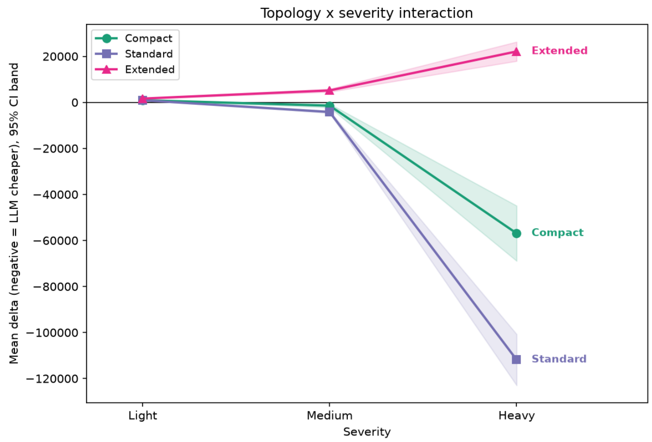


**Figure 1:** Phase 2: mean $\Delta$ vs. severity, one line per topology (signed-sqrt axis; bands are 95% CI). Compact/Standard diverge negative (LLM-favorable); Extended diverges positive (heuristic-favorable).

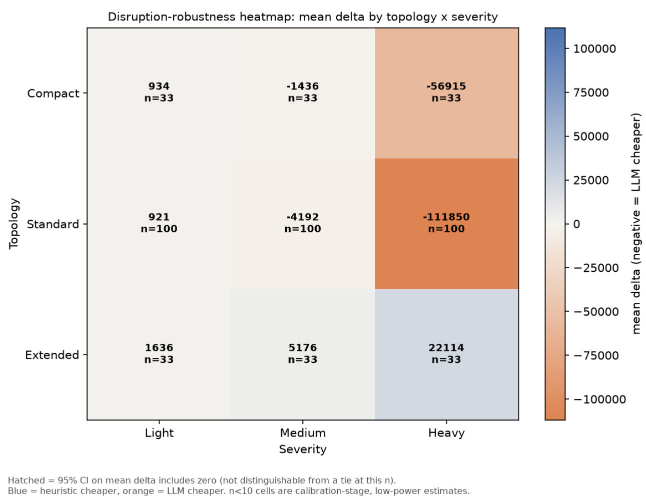


**Figure 2:** Phase 2: mean $\Delta$ across the full grid (diverging color; orange = LLM-favorable, blue = heuristic-favorable). Every cell's effect is statistically real at its sample size.

ure 2: no cell is hatched, since every 95% CI excludes zero, so every color shown is a real effect.

Separately, Compact × Medium becomes the first cell in either phase to land away from unanimity (75.8% LLM, 24.2% heuristic) – direct evidence the redesign produces genuine per-replication variation. Consistent with this, the signal-to-noise ratio is defined as:

$$\mathrm{SNR} = \frac{|\bar{\Delta}|}{\mathrm{sd}(\Delta)} \tag{4}$$

and it sits at 0.6×–3.0× across all nine Phase 2 cells, against the 7.6×–10.2× recomputed identically on Phase 1's real runs (Figure 3) – the audited root cause is measurably fixed, not merely asserted fixed.

# 6 Discussion: Synthesis Across Both Phases

Combining 798 replications across two phases and four network configurations, the study's answer to its research question

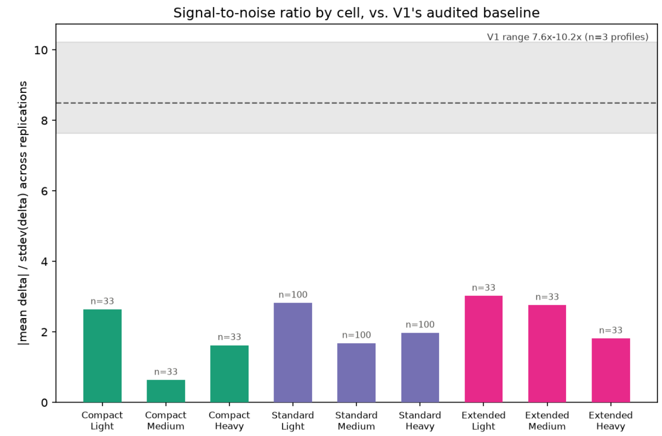


**Figure 3:** $|\bar{\Delta}|/\mathrm{sd}(\Delta)$ per Phase 2 cell against Phase 1's real, identically-computed baseline (shaded band). Every Phase 2 cell sits well under Phase 1's 7.6×–10.2×.

is conditional, not universal.

**Pros of the agentic policy.** Substantially larger cost reduction under severe disruption on constrained-to-moderate topologies (up to $-111{,}850$ mean $\Delta$); the only policy to ever produce a genuinely mixed, non-unanimous outcome (Compact × Medium), suggesting situational judgment rather than a fixed rule; this advantage replicated independently across two separate live-API runs on the same network (Phase 1 and Phase 2's Standard column).

**Cons.** Consistently worse under Light disruption in every configuration tested (`WAIT` is usually already optimal, and reasoning/tool-call overhead buys nothing); worse at every severity on the highest-redundancy topology; real monetary cost and ∼2.6 s of live-API latency per decision, against a free, instant, deterministic heuristic evaluation.

A plausible mechanism for the Extended reversal: the heuristic exhaustively scores every candidate route by estimated cost and never has to judge which of many plausible options is worth pursuing, while a bounded agent (max 8 tool calls) reasons under genuinely higher ambiguity as the option count grows – plausible, not proven (see Limitations).

# 7 Limitations and Nuance

Four nuances qualify the combined findings.

(1) Extended's severity scenarios target `hub_1`, not `port_primary` – justified (Section 5), but it entangles the reversal with *which node* is attacked, not topology redundancy alone.

(2) Standard's shipment quantity stays fixed across both phases by design, while Compact/Extended draw real variance – an acknowledged confound on any comparison involving Standard specifically.

(3) Phase 2's 33 replications/cell is real, CI-confirmed power, but less than Phase 1's 100/profile; Compact × Medium's split could plausibly shift with more replications.

(4) Both phases use one model, one temperature, one prompt template, and one narrow decision (single-shipment routing) – this is a finding about one policy design tested twice, not a general claim about agentic AI in logistics.

# 8 Follow-up Directions

## 8.1 Relaxing Scope Simplifications

Before turning to a fundamentally different policy class, several scope simplifications made deliberately for this study's isolation of concerns are natural nearer-term extensions of the same paired-simulation methodology, requiring no change to the fairness mechanism itself (Section 3).

**Multiple products.** Both phases model exactly one product moving through the network; a shipment is a scalar quantity, never a composition of distinct goods. Extending to multiple products sharing the same nodes and edges would introduce genuine resource contention – a rerouted shipment of product A can crowd out product B's capacity – that a single-product simulator cannot express by construction, letting decision quality be tested under a richer, more realistic notion of scarcity.

**Supplier selection and procurement.** The network's supplier is fixed and singular; neither policy ever chooses *where* to source from, only how to move what is already committed. Introducing multiple candidate suppliers with their own cost/reliability trade-offs would test a qualitatively different decision – procurement, not routing – on the same underlying infrastructure.

**Shipment splitting and joint multi-shipment optimization.** Every decision here is scoped to exactly one shipment in isolation; a policy cannot split a shipment across routes or jointly optimize several shipments' decisions together, even when doing so would be jointly cheaper. Both restrictions were deliberate – they keep the decision unit simple enough to validate exhaustively – but relaxing either tests whether either policy's advantage survives a combinatorially larger action space.

**Dynamic pricing and fleet-level routing.** Transport costs and capacities are fixed per edge for a given day; nothing here models a carrier's own pricing response to demand, or routes a fleet of vehicles rather than abstract edge capacity. Both are standard extensions in classical logistics optimization and would test the same two policy types against a substantially richer cost surface.

Each of these is an extension to *what the environment models*, in the same spirit as Phase 2's own topology and randomness extensions to Phase 1 – not a change to how either policy is scored or compared.

## 8.2 A Reinforcement-Learning Formulation

Both policies studied here are, in a precise sense, *not learned*: the heuristic is a fixed rule, and the LLM agent's behavior comes from pre-trained general reasoning plus a prompt, not from experience inside this environment. A longer-term follow-up is to add a third arm that *is* learned: a reinforcement-learning (RL) policy trained directly against the simulator's own state, action, and reward structure.

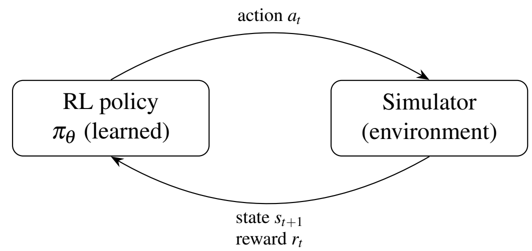


**Figure 4:** The RL formulation reuses the state/action/reward surface both existing policies already share: the same `DecisionObservation` as state $s_t$, the same four-choice action $a_t \in \{$`WAIT`, `REROUTE`, `EXPEDITE`, `ABSTAIN`$\}$, and $r_t = -J_t$ (Eq. 1) as reward.

The formulation is close to already sitting in the codebase (Figure 4):

**State.** The same read-only `DecisionObservation` both existing policies already receive – shipment facts, destination and shock context, current plan, candidate routes.

**Action.** The same four-choice space, `WAIT/REROUTE/EXPEDITE/ABSTAIN`, already shared and validated identically.

**Reward.** The negative of the same daily cost $J$ (Eq. 1) already accumulated for TCD, either shaped per-day (dense) or given once at episode end as $-$TCD (sparse, but exactly the metric this study already optimizes for by construction).

Because the simulator's transitions are already deterministic and the event tapes are already fully pre-generated and replayable, an RL agent can be trained across thousands of simulated episodes at effectively zero marginal cost – a sharply different economics from the LLM arm, where every decision is a paid, $\sim$2.6 s API call, and one of the more interesting comparisons this study does not yet make: cost-per-decision at training time versus cost-per-decision at deployment time.

What it could add beyond the current two-policy comparison:

(i) **An empirical reference point.** If a policy free to learn via trial and error over many episodes converges to a similar decision boundary as the heuristic or the LLM, that is independent evidence neither current policy is leaving much value on the table; if it instead discovers materially different behavior (for instance, rerouting proactively before a shock is fully disclosed, something neither current policy is built to do), that is itself a finding.

(ii) **A way to separate environment from agent** in the Extended-topology reversal reported above: if a trained RL policy *also* reverses on Extended, that argues the reversal is a structural property of the decision problem itself, independent of which agent is solving it; if an RL policy does not reverse there, that argues the LLM's specific bounded-reasoning behavior, not the environment, is responsible – a distinction Section 7 could not resolve with the two policies tested here.

This is not a free upgrade. An RL policy loses the heuristic's auditability and the LLM's natural-language reasoning trace – its decision boundary is a learned function, not a rule a person can read. Reward shaping is itself a research decision with failure modes (a dense per-day reward can encourage short-sighted behavior; a sparse terminal reward is harder to learn from). And a policy trained on one topology/severity distribution is not guaranteed to generalize to another without deliberately training across the full nine-cell grid, which reintroduces, at training time, exactly the topology-dependence this study's headline result already shows matters.

# 9 Conclusion

Two phases, one shared paired-simulation infrastructure, and 798 live-API replications give a conditional but well-evidenced answer to whether agentic AI mitigates supply-chain disruption better than a classical heuristic: **it depends on both disruption severity and network structure, and the dependence on structure is not monotonic in the way Phase 1 alone would have suggested.** The agent's advantage is real and large on constrained-to-moderate networks under severe disruption; the heuristic's advantage is equally real, and grows rather than shrinks, on the most redundant network tested. Phase 1's audit-driven redesign into Phase 2 is itself part of the finding: an experiment that cannot produce anything but unanimous agreement is not evidence of robustness, and the discipline of building an environment able to disagree with itself is what surfaced the topology-reversal result in the first place. Near-term, future work should decouple Extended's shock target from its topology, equalize shipment-volume variance across tiers, extend the demand-side and supplier-side shock types already built but not yet run at grid scale, and relax the single-product and single-supplier scope simplifications discussed above (Section 8.1); longer-term, the reinforcement-learning formulation (Section 8.2) offers a way to test whether either policy studied here is close to a ceiling, or whether the topology-reversal finding is about the environment rather than the agent.

# References


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